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React JS Lesson 14 – Context API (Premium Notes)

1) What is Context API

- Context API is a React feature used to share data globally between components.
- It avoids repeated prop passing.

Example:

```
const UserContext = createContext();
```

Explanation:

- createContext creates shared data container.

2) Why Context API is needed

- Needed when many components require same data.
- Useful for theme, login user, language.

Example:

Theme, User Data, Cart Data

Explanation:

- One source can serve many components.

3) Props Drilling Problem Explained

- Without Context API, props travel through unnecessary components.

Example:

Parent → Child1 → Child2 → Child3

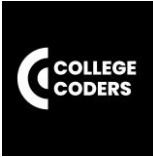
Explanation:

- Middle components only forward data.

4) createContext in React

- createContext creates context object.

Example:



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```
import { createContext } from 'react';
```

```
const UserContext = createContext();
```

Explanation:

- Context object must be exported if reused.

5) Context.Provider usage

- Provider supplies data to child tree.

Example:

```
<UserContext.Provider value={name}>  
  <Child />  
</UserContext.Provider>
```

Explanation:

- All children inside provider can access value.

6) Provider Scope concept

- Only components inside provider can access context.

Example:

Inside Provider → access allowed
Outside Provider → no access

Explanation:

- Scope matters.

7) useContext Hook

- useContext reads context data directly.

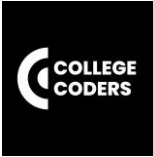
Example:

```
const user = useContext(UserContext);
```

Explanation:

- Current value is received instantly.

8) Selective Data Access



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- Only component needing data reads it.

Example:

```
function Child(){  
  const user = useContext(UserContext);  
}
```

Explanation:

- Middle components can be skipped.

9) Difference between Props Drilling vs Context API

- Props drilling passes step by step.
- Context gives direct access.

Example:

Props → multiple forwarding

Context → direct reading

Explanation:

- Context reduces complexity.

Mini Project – User Name Using Context API

- Create UserContext.
- Provide name from parent.
- Read directly in child.